MILITARY SPECIFICATION

PROPELLANT, SOLVENTLESS, TYPE M37 FOR PROJECTILE, 155MM, HE, ROCKET ASSISTED, M549 AND M549A1

This amendment forms a part of Military Specification MIL-P-63201, dated 13 September 1983 and is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all departments and agencies of the Department of Defense.

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3.2, TABLE I:

Change: "nitroglycerine, Type F" to "nitroglycerine, Type 1".

Change: "2 Nitrodipheneylamine MIL-N-399" to "2 Nitrodiphenylamine MIL-N-3399".

3.3: Delete in its entirety and substitute the following:

"3.3 Strand burning rate. The average of five individual strand burning rate determinations at each temperature and pressure shall meet the requirements of Table II. If one or more of the average burning rates fail the requirements of Table II, the sum of the squares of the differences shall not exceed 0.0007 inch squared per second squared.

TABLE II
STRAND BURNING RATE

Pressure Pounds per Square Inch	Burning Rate Inches per Second	
	Lower Limit @ -40°F	Upper Limit @ +140°F
600	.359	.450
800	.414	.500
1000	.433	.475
1200	.368	.462

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Pressure Pounds per Square Inch	Burning Rate Inches per Second	
	Lower Limit @ -40°F	Upper Limit @ +140°F
1400		.479
1600		.581
1800		.615
2000		.663

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4.4.1: Delete in its entirety and substitute the following:

"4.4.1 Propellant testing.

4.4.1.1 Sampling. Each lot of propellant submitted for acceptance shall be sampled by the contractor under the direction of the Government inspector. Twenty five individual even-speed-rolled sheets shall be selected at random for each inspection lot (see 6.3). A sample at least 11 inches long by 4 inches wide shall be cut at random from each of the 25 sheets selected. The long axis of each sample shall be in a direction which is parallel to the direction of final rolling. The samples shall be numbered consecutively to facilitate proper testing sequence. From each of the samples above, test samples shall be selected and prepared as described in 4.4.1.2 through 4.4.1.5 and Figure 1.

4.4.1.2 Test Sample A. Heat of explosion determination. Eight (8) samples shall be selected at random from the 25 samples designated. From each of the eight samples, a portion weighing at least 20.0 grams shall be reduced in size as described in Method 509.3 of MIL-STD-286 to pass through a U.S. No. 20 mesh screen. Each of the eight resulting samples shall be stored in a separate moisture-tight container.

4.4.1.3 Test Sample B. Strand burning rate determination. To obtain a total of 60 acceptable strands, two acceptable strands shall be removed from 15 of the original 25 samples and three acceptable strands shall be removed from the remaining 10 samples. The strands shall be of the width prescribed in Method 803.1 of MIL-STD-286. Since the strands will be too long, they shall be cut to the correct length range. The residual short lengths shall be used for Test Sample C. The long strands for Sample B shall be stored in moisture tight containers and properly identified.

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4.4.1.4 Test Sample C. Moisture determination. 25 moisture samples representing each of the 25 original samples shall be selected from the residual short lengths from the preparation of Sample B. The selected moisture samples shall be cut to a length of approximately 3.0 inches. These 3.0 inch samples shall be diced into 1/8 inch squares. All resultant squares shall be thoroughly blended together, stored in a moisture-tight container and properly identified. At least 20 grams of propellant will be required.

4.4.1.5 Test Sample D. General chemical analysis and taliani test. From the material remaining after the preparation of samples A, B and C a section weighing at least 4.0 grams shall be selected from each of the original 25 samples. These sections shall be reduced in size as described in Method 509.3, MIL-STD-286 and blended thoroughly to give a uniform sample weighing at least 160 grams. The sample shall be stored in a moisture-tight container and properly identified."

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4.5.1: Delete the first sentence and substitute the following:

"To determine conformance of the propellant with 3.2, 3.4, 3.5 and 3.6 the following analysis shall be carried out on the samples taken in 4.4.1.2, 4.4.1.4 and 4.4.1.5.

TABLE III:

Change: "228.1" to "T228.1".

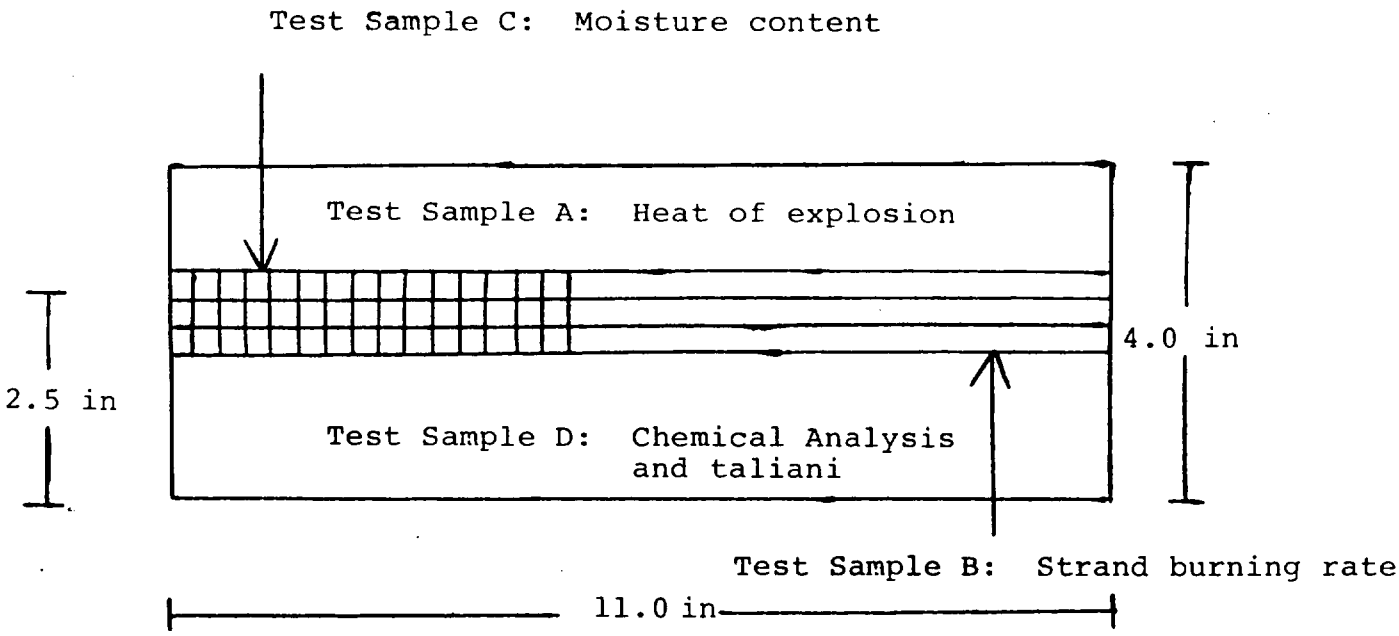
Change: "225.1" to "T225.1".

4.5.2: Change "25" to "60".

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FIGURE 1

DIVISION OF SHEET SAMPLE INTO
FOUR INDIVIDUAL TEST SAMPLES



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